Netflix-Business Case Analysis

Submitted by:A Shreenidhi **Batch**:DSML Dec25(2) Beginner(MWF Night)

Context:

```
Netflix is one of the world's leading media and video streaming platforms, c
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
url="https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/000/940/
```

```
df=pd.read_csv(url)
```

1.Problem Statement & Basic Metrics: Netflix wants to strengthen its content strategy and grow its subscriber base across countries. This analysis explores the Netflix catalog (movies and TV shows, along with attributes like cast, director, country, genre, rating, and release year) to identify:

What type of content (movies vs. TV shows) Netflix should focus on producing Which genres and content types perform well in which countries/regions Trends over time in content addition and release Recommendations for content strategy and market expansion

```
df.head(3)
```

	show_id	type	title	director	cast	country	date_added	releas
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel	NaN	September 24, 2021	

Next steps: New interactive sheet

```
#Rows and columns
df.shape
```

```
(8807, 12)
```

```
#unique values per column
df.nunique()
```

	0
show_id	8807
type	2
title	8807
director	4528
cast	7692
country	748
date_added	1767
release_year	74
rating	17
duration	220
listed_in	514
description	8775

dtype: int64

Totally 8807 unique movie/shows altogether. There are missing values in the column director,case,country,date_addded,release_year,rating and duration series.

```
df['date_added'].head(1)
```

	date_added
0	September 25, 2021

dtype: object

```
df['date_added'].tail(1)
```

	date_added
8806	March 2, 2019

dtype: object

The first day and last day on which the movie/show is added is 02/03/2019 and 25/09/2021

2. Shape, Data Types, Category Conversion, Missing Values, Statistical Summary

```
#a.Shape
print("Shape:", df.shape)
```

```
Shape: (8807, 12)
```

Dataset has 8807 rows and 12 columns

```
#b.Data types
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
#c.Convert appropriate columns to 'category'
df['type'].astype('category')
df['rating'].astype('category')
```

	rating
0	PG-13
1	TV-MA
2	TV-MA
3	TV-MA
4	TV-MA
...	...
8802	R
8803	TV-Y7
8804	R
8805	PG
8806	TV-14

8807 rows × 1 columns

dtype: category

Converted type and rating.

```
# d.Missing values
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

Director column has highest missing values comparing to other columns.

```
#Statistical summary
df.describe()
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

The data shows that Netflix's catalog is skewed toward recent content, with a median release year of 2017 and 75% of titles released in or after 2019, even though the range goes back to 1925. Movies make up the majority of the catalog at about 70% (6,131 titles), while TV Shows form a smaller share, indicating Netflix's current library leans more toward movies than shows.

3. Non-Graphical Analysis: Value Counts and Unique Attributes

```
#value counts
df["country"].value_counts()
```

country	count
United States	2818
India	972
United Kingdom	419
Japan	245
South Korea	199
...	...
Mexico, United States, Spain, Colombia	1
Canada, Norway	1
Finland, Germany, Belgium	1
Argentina, United States, Mexico	1
United Kingdom, United States, Germany, Denmark, Belgium, Japan	1

748 rows × 1 columns

dtype: int64

```
df['type'].value_counts()
```

type	count
Movie	6131
TV Show	2676

dtype: int64

```
uni=df[['country','type']]
print(uni)
```

```
           country     type
0    United States    Movie
1     South Africa  TV Show
2              NaN  TV Show
3              NaN  TV Show
4            India  TV Show
...            ...      ...
8802 United States    Movie
8803           NaN  TV Show
8804 United States    Movie
8805 United States    Movie
8806         India    Movie

[8807 rows x 2 columns]
```

```
uni.head(10)
```

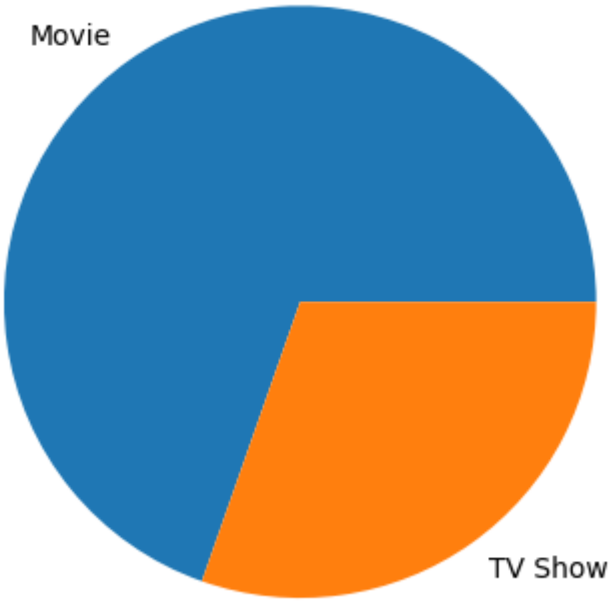
	country	type
0	United States	Movie
1	South Africa	TV Show
2	NaN	TV Show
3	NaN	TV Show
4	India	TV Show
5	NaN	TV Show
6	NaN	Movie
7	United States, Ghana, Burkina Faso, United Kin...	Movie
8	United Kingdom	TV Show
9	United States	Movie

Next steps: New interactive sheet

```
count_t=df["type"].value_counts()
print(count_t)
```

```
type
Movie      6131
TV Show    2676
Name: count, dtype: int64
```

```
plt.pie(count_t,labels=count_t.index)
plt.show()
plt.figure(figsize=(3,1))
```




```
<Figure size 300x100 with 0 Axes>
<Figure size 300x100 with 0 Axes>
```

This is the value counts of movies and Tv shows from all countries in the data

```
movie_list_top10=uni[uni["type"]=="Movie"].value_counts().head(10)
movie_list_top10
```

country	type	count
United States	**Movie**	2058
India	**Movie**	893
United Kingdom	**Movie**	206
Canada	**Movie**	122
Spain	**Movie**	97
Egypt	**Movie**	92
Nigeria	**Movie**	86
Indonesia	**Movie**	77
Japan	**Movie**	76
Turkey	**Movie**	76

dtype: int64

Top 10 countries in terms of Number of movies are United States,India,United Kingdom,Canada,Spain,Egypt,Nigeria,Indonesia,Turkey,Japan

4.Visual Analysis - Univariate, Bivariate after pre-processing of the data

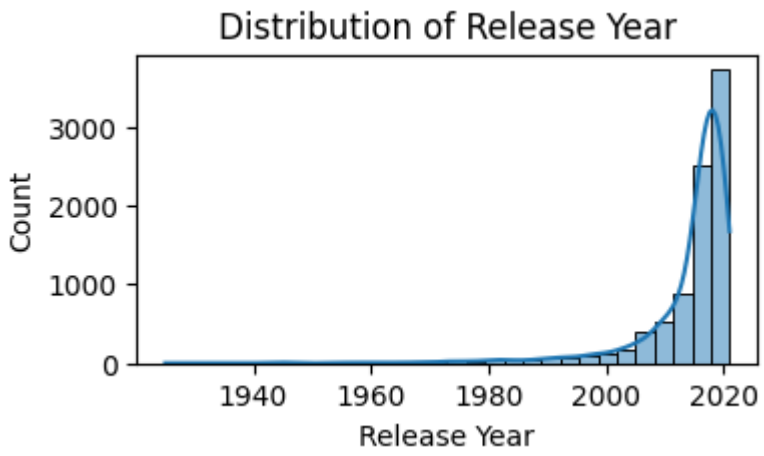
Unnesting the columns

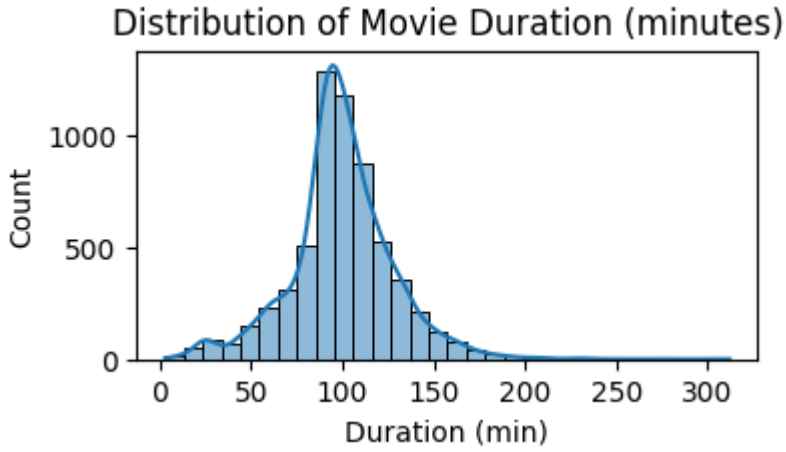
```
# Unnest Country
df_country = df.assign(country=df['country'].str.split(', ')).explode('count
# Unnest Cast
df_cast = df.assign(cast=df['cast'].str.split(', ')).explode('cast')
# Unnest Director
df_director = df.assign(director=df['director'].str.split(', ')).explode('di
# Unnest Genre
df_genre = df.assign(listed_in=df['listed_in'].str.split(', ')).explode('lis
# Split duration into numeric value + unit (movies=min, TV shows=seasons)
df['duration_num'] = df['duration'].str.extract(r'(\d+)').astype(float)
df['duration_type'] = df['duration'].str.extract('([a-zA-Z]+)')
```

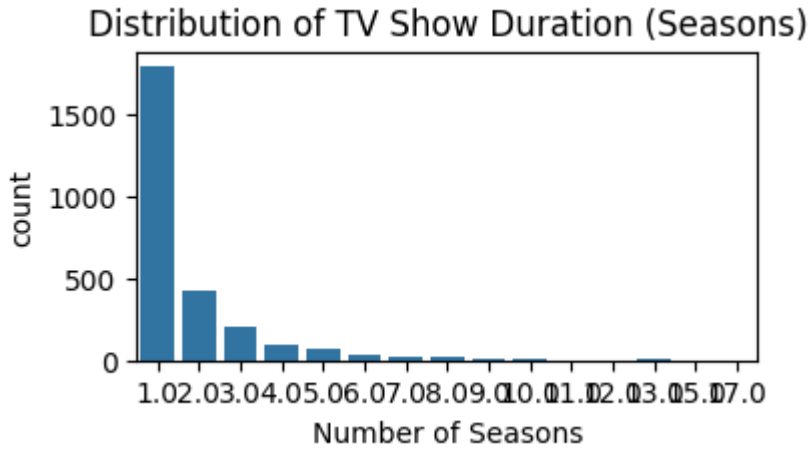
```
#a.Continuous Variable
# Release year - histogram
plt.figure(figsize=(4,2))
sns.histplot(df['release_year'], bins=30, kde=True)
plt.title('Distribution of Release Year')
plt.xlabel('Release Year')
plt.ylabel('Count')
plt.show()

# Duration for Movies - histogram
plt.figure(figsize=(4,2))
sns.histplot(df[df['type']=='Movie']['duration_num'], bins=30, kde=True)
plt.title('Distribution of Movie Duration (minutes)')
plt.xlabel('Duration (min)')
plt.show()

# Duration for TV Shows (seasons) - countplot since it's discrete/small rang
plt.figure(figsize=(4,2))
sns.countplot(x='duration_num', data=df[df['type']=='TV Show'])
plt.title('Distribution of TV Show Duration (Seasons)')
plt.xlabel('Number of Seasons')
plt.show()
```



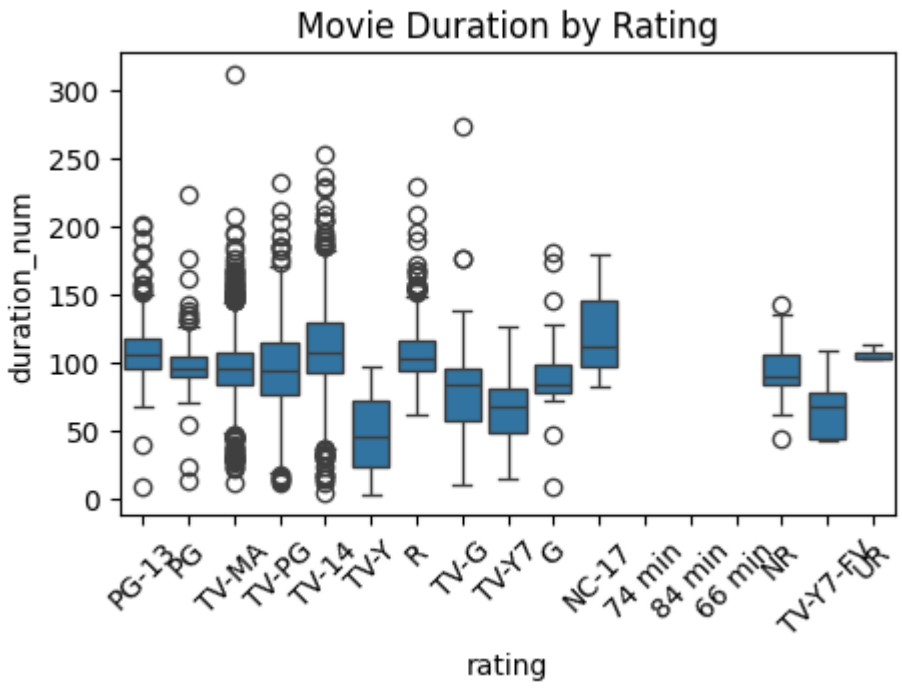


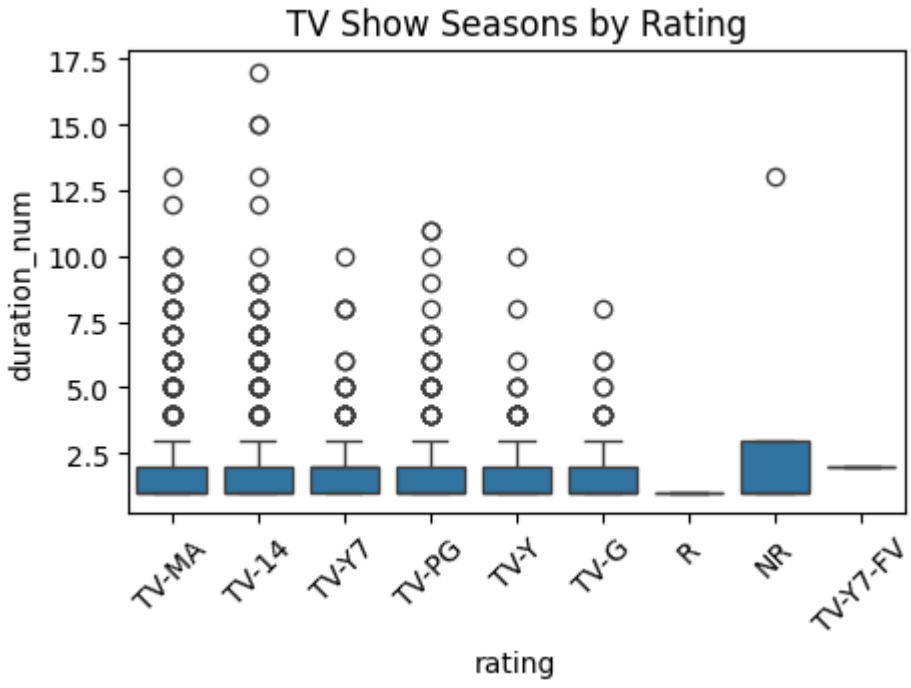


Release year skews heavily recent (post-2015), movie durations cluster normally around 90-110 minutes with a long tail of outliers, and TV show seasons are extremely right-skewed with most shows having just 1 season. Movies dominate the catalog over TV Shows at roughly a 70:30 split, though this alone doesn't confirm Netflix's current production strategy

```
#b.Categorical Variable
# Movie duration by rating
plt.figure(figsize=(5,3))
sns.boxplot(x='rating', y='duration_num', data=df[df['type']=='Movie'])
plt.xticks(rotation=45)
plt.title('Movie Duration by Rating')
plt.show()
```

```
# TV Show seasons by rating
plt.figure(figsize=(5,3))
sns.boxplot(x='rating', y='duration_num', data=df[df['type']=='TV Show'])
plt.xticks(rotation=45)
plt.title('TV Show Seasons by Rating')
plt.show()
```





Movie duration by rating: Median durations stay fairly consistent (85-115 min) across most ratings, though mature categories like TV-14 and R show more high-duration outliers.

TV show seasons by rating: Nearly all ratings have a median of just 1-2 seasons, with TV-14, TV-MA, and TV-PG showing the most outliers of long-running shows (10+ seasons).

5. Missing Value & Outlier Check

```
#missing value
df.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0
duration_num	3
duration_type	3

dtype: int64

Director attribute has the highest missing values.

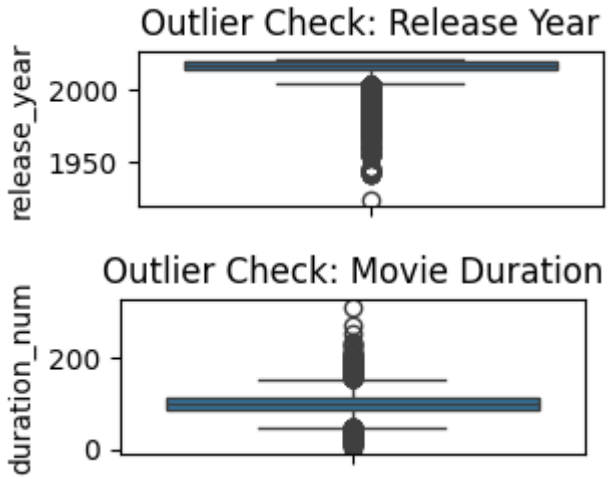
```
# Outlier check using IQR method
def check_outliers(col):
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    outliers = df[(df[col] < lower) | (df[col] > upper)]
    print(f"{col}: Lower={lower}, Upper={upper}, Outliers={len(outliers)}")
    return outliers
```

```
check_outliers('release_year')
check_outliers('duration_num')

# Visual check
plt.figure(figsize=(3,1))
sns.boxplot(y=df['release_year'])
plt.title('Outlier Check: Release Year')
plt.show()

plt.figure(figsize=(3,1))
sns.boxplot(y=df[df['type']=='Movie']['duration_num'])
plt.title('Outlier Check: Movie Duration')
plt.show()
```

```
release_year: Lower=2004.0, Upper=2028.0, Outliers=719
duration_num: Lower=-154.0, Upper=262.0, Outliers=2
```



Old release years and long/short movie durations are natural outliers, not data errors, so no treatment is needed.

6. Insights based on Non-Graphical and Visual Analysis

```
#a.Comments on the range of attributes
df['release_year'].describe()
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

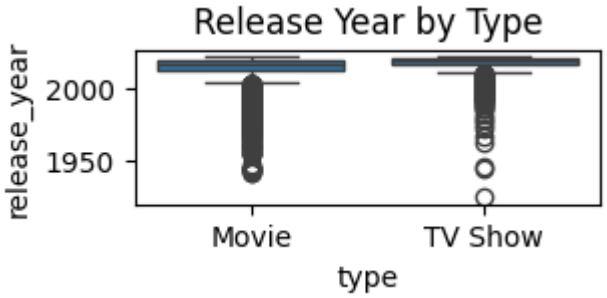
dtype: float64

Comments: release_year ranges from 1925 to 2021, but most content (75%) is from 2013 onwards, showing Netflix's catalog is mostly recent titles. The mean (2014) and median (2017) being close to the max year confirms the data is skewed toward newer releases rather than spread evenly across time..

```
#b.Comments on the distribution of the variables and relationship between th
print("\nMedian release year by type:")
print(df.groupby('type')['release_year'].median())

plt.figure(figsize=(3,1))
sns.boxplot(x='type', y='release_year', data=df)
plt.title('Release Year by Type')
plt.show()
```

```
Median release year by type:
type
Movie      2016.0
TV Show    2018.0
Name: release_year, dtype: float64
```



Comments: `release_year` ranges from 1925 to 2021, but most content (75%) is from 2013 onwards, showing Netflix's catalog is mostly recent titles. The mean (2014)

and median (2017) being close to the max year confirm this skew toward newer

```
#c.Comments for each univariate and bivariate plot
```

Comment: Univariate:

Release Year (distplot): Sharp concentration post-2015; catalog skews recent. Movie Duration (distplot):Near-normal distribution centered at ~90-110 min, with few long outliers (200-300 min). TV Show Seasons (countplot): 1,800 shows have exactly 1 season; count drops sharply beyond that — most shows aren't renewed multiple times.